

# Team Progress Update #1

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MIME-5504A



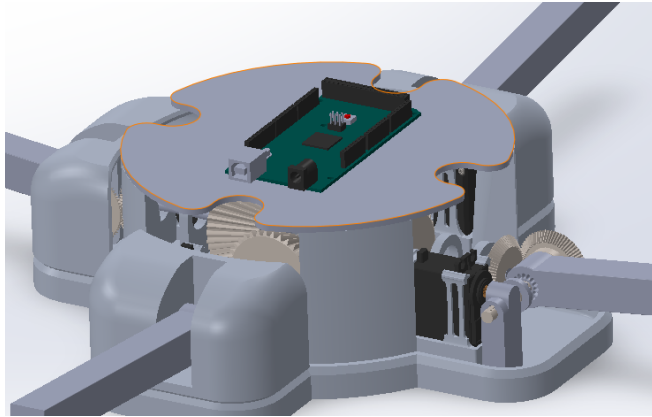
Spring 2025 Accomplishments:

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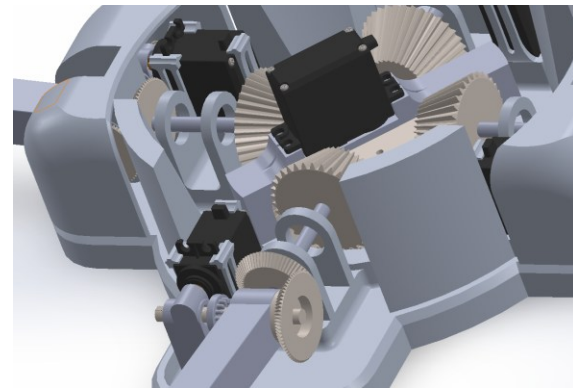
# CAD Prototype 1: Black Box

## CAD Progress:

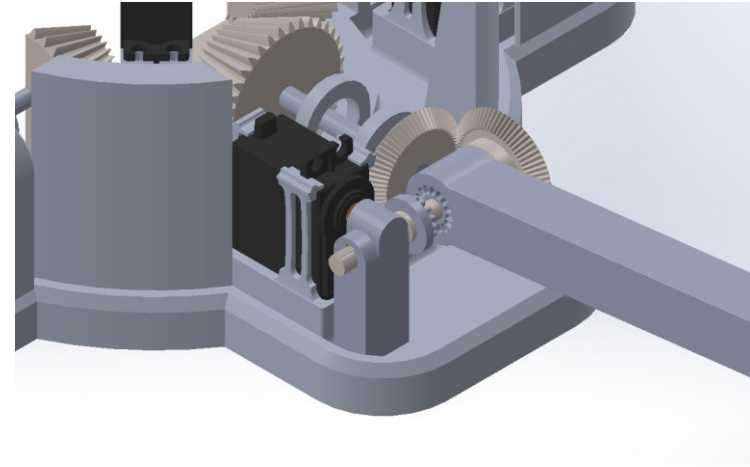
- Switchboard
- Gear Development
- McMaster-Carr BOM Inclusions



Black Box Module of Attachment

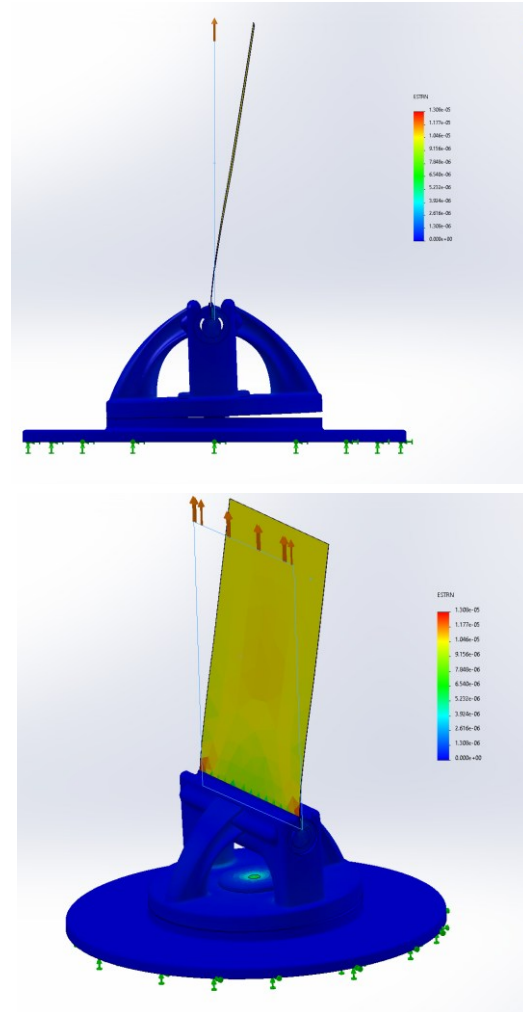
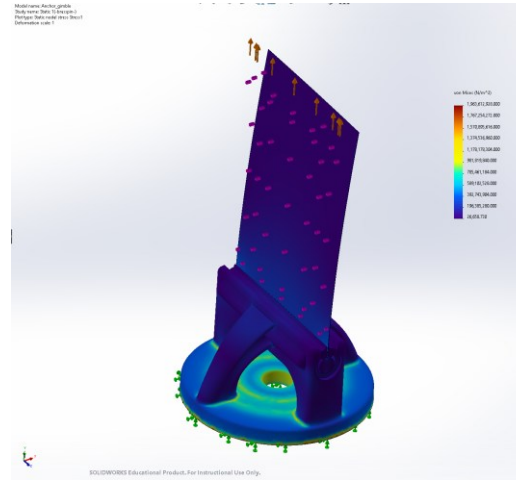
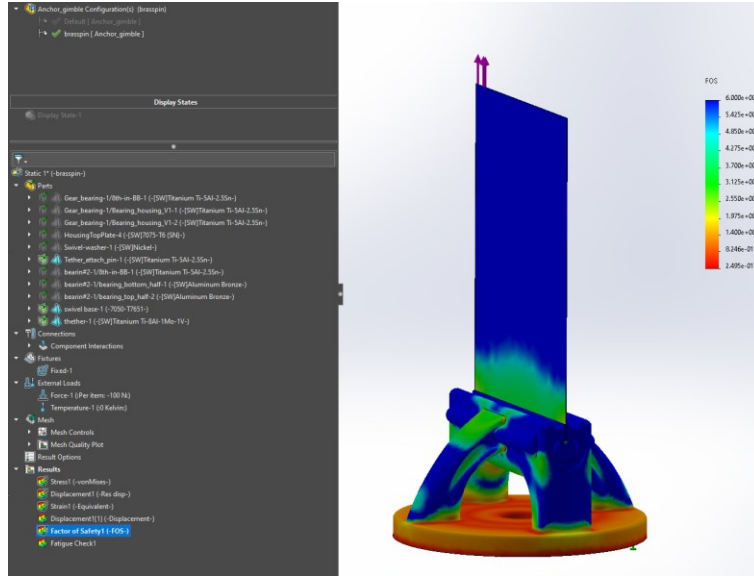


Gear Functionality Exploded View



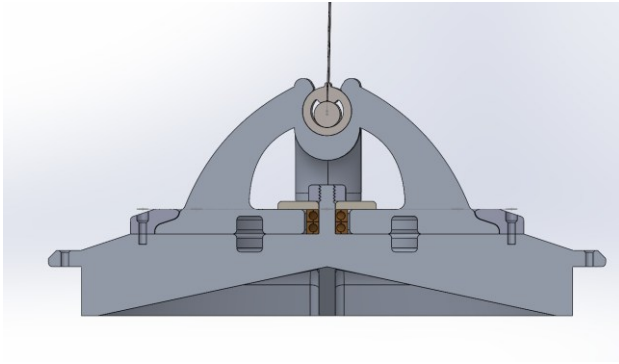
Cross-Sectional View of Black Box Module

# FEA of Tether Gimbal Attachment Point - Highlights

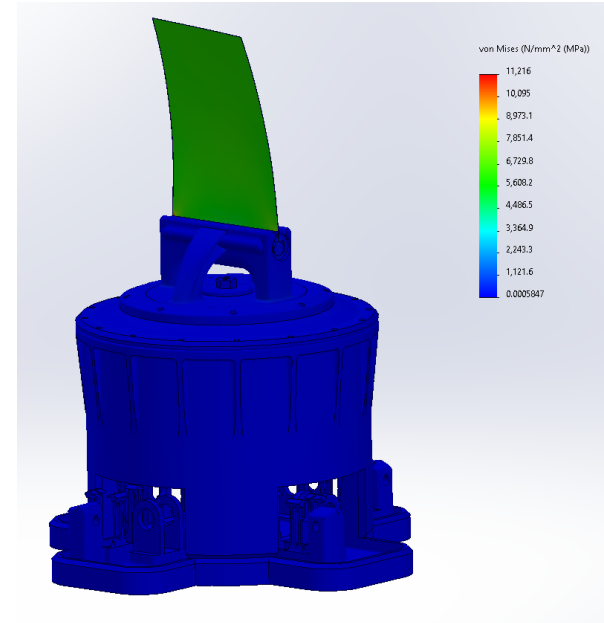
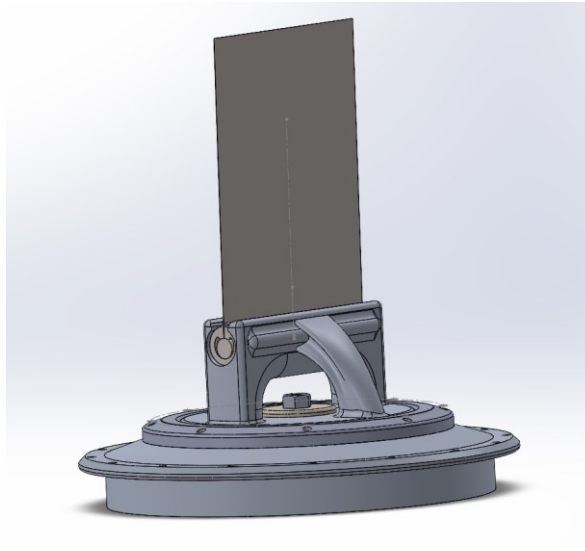


| CR# | CR Description                                                                             | Weight | Matching Engineering Specification                                                            |
|-----|--------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------|
| 6   | Strong enough material to perform mandatory functions & withstand varied impact conditions | 10     | Shear Strength $\geq 205$ MPa, UTS $\geq 600$ MPa, Yield Strength $\geq 385$ MPa (Al 6061-T6) |
| 9   | Ability to effectively interface with tether device                                        | 5      | Attachment point to relevant team concept                                                     |

# FEA Influenced Redesign



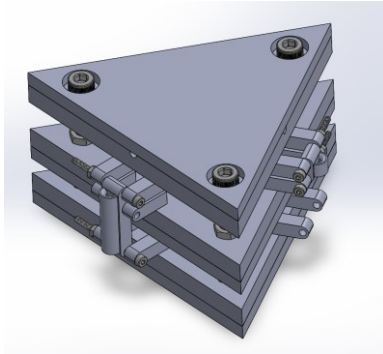
- Gimbal Constraining Ring
- Plate Support Ribbing
- Black Box Shell Support Ribbing



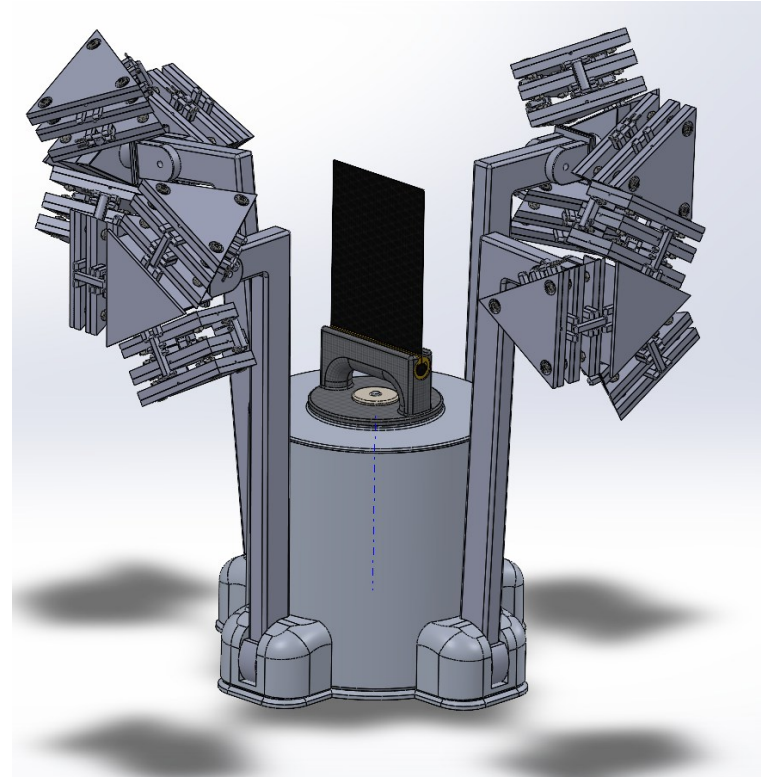
# CAD Prototype 2: Feet

## CAD Key Functions:

- Folding Mechanism
- Wire for expanding DoF
- McMaster-Carr BOM Inclusions



Unfolding Foot Module



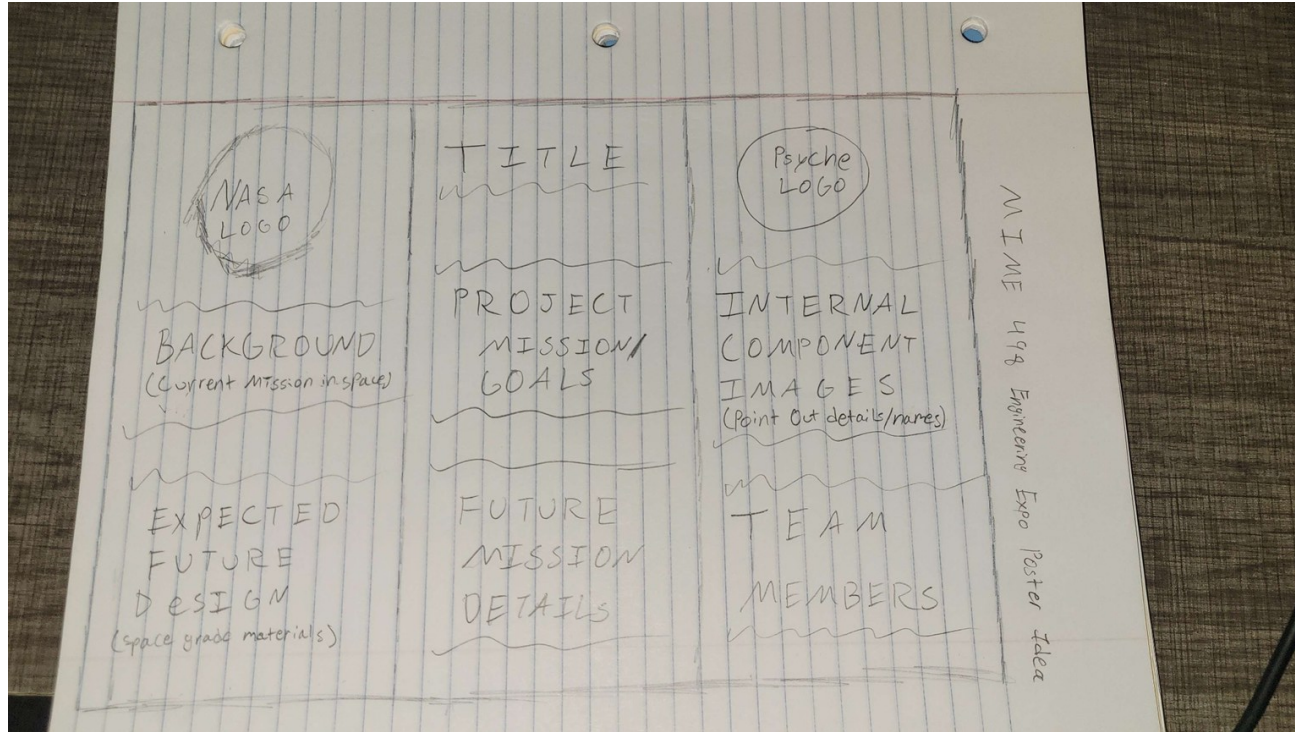
Overall Assembly With Feet and Legs

| CR# | CR Description                                        | Weight | Matching Engineering Specification                                                                                   |
|-----|-------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------|
| 5   | Able to self-anchor in near-zero gravity environments | 10     | Is able to land and anchor while experiencing <1% of Earth's gravity (-0.106 m/s <sup>2</sup> minimum)               |
| 8   | Ability to anchor with an uneven surface              | 15     | Anchors can attach onto surfaces that have height differences of up to 10 meters, handle at least 34 degrees of tilt |

# MBOM Progress & Purchases

| Item Name                                                                                                                  | Brand             | Quantity | Unit Cost                | Total Cost | Price Checked With Receipt | Source                             | Date Purchased (Requested) | Date Received |
|----------------------------------------------------------------------------------------------------------------------------|-------------------|----------|--------------------------|------------|----------------------------|------------------------------------|----------------------------|---------------|
| 12V Power Supply                                                                                                           | ALITOVE           | 1        | \$9.58                   | \$9.58     | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 51110 Thrust Ball Bearings 50mm x 70mm x 14mm Chrome Steel ABEC3 Single Row Roller 2pcs                                    | uxcell            | 1        | \$13.19                  | \$13.19    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 20-Pack R4-2RS Bearings, ID 1/4 x OD 5/8 x 0.196 inch Small Ball Bearings                                                  | Donepart          | 1        | \$15.99                  | \$15.99    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 320pcs M3 Circuit Board Mounting Kit                                                                                       | Lystali           | 1        | \$9.99                   | \$9.99     | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 2 Pack, 1" Ball Adapter with M10 x 1.25 x 15 Threaded Post Compatible with RAM Mounts B Size 1 Inch Ball Double Socket Arm | BROOVAN           | 2        | \$17.14                  | \$34.28    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 22 awg Silicone Electrical Wire (100ft)                                                                                    | Haerkin           | 1        | \$16.98                  | \$16.98    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| Pro 950 Polyimide Film Tape, 7500V Dielectric Strength, 36 yds Length x 1" Width                                           | Pro Tapes         | 1        | \$13.56                  | \$13.56    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| [4-Pack] MG996R 55g Servo Motors                                                                                           | Deegoo FPV        | 1        | \$17.99                  | \$17.99    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 40KG Digital Servo Motor 360 Degrees                                                                                       | DIYmail           | 1        | \$28.99                  | \$28.99    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 10pcs High Voltage Generator DC 6-12V to 1000KV 1000000V                                                                   | XIITIA            | 1        | \$24.79                  | \$24.79    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| Epoxy Resin Kit 72OZ                                                                                                       | Puduo             | 1        | \$36.99                  | \$36.99    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 99.9% Solid Bare Copper Wire 18 Gauge                                                                                      | YEZHET            | 1        | \$20.45                  | \$20.45    | No                         | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/22/2025     |
| SN74HC595N IC 8-Bit Shift Registers                                                                                        | Texas Instruments | 1        | \$7.35                   | \$7.35     | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| ELEGOO MEGA R3 Board (Arduino Mega Board)                                                                                  | ELEGOO            | 1        | \$21.84                  | \$21.84    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| 5PCS GY-521 6 DOF MPU6050 Accelerometer Sensor Module                                                                      | Stemedu           | 1        | \$13.70                  | \$13.70    | Yes                        | <a href="#">Amazon Link</a>        | 4/4/2025                   | 4/9/2025      |
| M5 x 0.8 mm Thread, 12 mm Long Bolt (50 Count)                                                                             | McMaster-Carr     | 1        | \$5.05                   | \$5.05     | No                         | <a href="#">McMaster-Carr Link</a> | 4/15/2025                  | 4/21/2025     |
| Class 04, M5 x 0.8 mm Thread Medium-Strength Steel Thin-Profile Hex Nut (100 Count)                                        | McMaster-Carr     | 1        | \$3.78                   | \$3.78     | No                         | <a href="#">McMaster-Carr Link</a> | 4/15/2025                  | 4/21/2025     |
| M18 x 2.5 mm Thread, 45 mm Long Alloy Steel Socket Head Screw Black-Oxide (5 Count)                                        | stemedu           | 6        | \$12.76                  | \$76.56    | No                         | <a href="#">McMaster-Carr Link</a> | 4/15/2025                  | 4/21/2025     |
| PETG 3D Printer Filament 1.75mm, (2.2lbs), (Orange)                                                                        | OVERTURE 3D       | 2        | \$21.99                  | \$43.98    | No                         | <a href="#">Amazon Link</a>        | 4/25/2025                  |               |
| PETG 3D Printer Filament 1.75mm, (4.4lbs), (Black (2-Pack))                                                                | OVERTURE 3D       | 2        | \$29.99                  | \$59.98    | No                         | <a href="#">Amazon Link</a>        | 4/25/2025                  |               |
|                                                                                                                            |                   |          | <b>Total Spent:</b>      | \$475.02   |                            |                                    |                            |               |
|                                                                                                                            |                   |          | <b>Total:</b>            | \$475.02   |                            |                                    |                            |               |
|                                                                                                                            |                   |          | <b>Remaining Budget:</b> | \$24.98    |                            |                                    |                            |               |

# Poster Mockups





# Poster Mockups

## COLLEGE OF ENGINEERING



### Background

NASA's mission to explore the metal-rich asteroid 16 Psyche represents one of the most ambitious exploration projects in recent space history. Psyche is thought to be the exposed core of an early planet, potentially composed of up to 90% metal (iron, nickel). Located in the asteroid belt between Mars and Jupiter, this unique celestial body offers unprecedented research opportunities. The mission was launched in October 2023, with arrival expected in 2026. The extreme environment of Psyche presents significant engineering challenges with minimal gravity (0.05-0.13 m/s<sup>2</sup>), extreme temperature variations (1-200K), and metallic surface composition requiring innovative attachment solutions.

### Fixture Design

- Space Grade Materials: High-temp composite material for feet, Aluminum 6061 structural components
- Radiation-hardened electronics and thermal protection
- Specialized dual-action anchoring (electrostatic + thermite bonding)
- Designed for extreme thermal cycling (1-200K)
- Adaptable feet to conform to uneven surfaces

### Operational Phases

- Transit Configuration: Legs folded for compact storage during journey
- Deployment Configuration: Legs extend and prepare for surface contact
- Anchored Configuration: Two-stage process:
  - Initial: Electrostatic adhesion provides temporary secure hold
  - Permanent: Thermite welding creates final bond



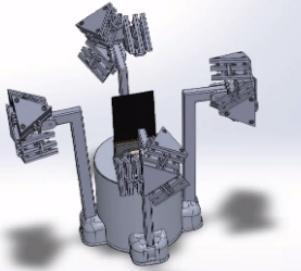
**Oregon State University**

## Mechanical, Industrial, and Manufacturing Engineering

# MIME-5504A: NASA Psyche Space Elevator - Attachment

### Project Mission:

Psyche is an asteroid with vast valuable resources, and we aim to develop an anchor for a space elevator to facilitate sample transfer between the asteroid and spacecraft. Our team's primary goal is to create a well-designed anchoring system that can effectively secure to Psyche's unique ferrous surface in microgravity conditions. The system must adapt to irregular terrain, withstand extreme temperature variations, and provide sufficient anchoring force to support space elevator operations.



### System Design Features

- Central Hub ("Black Box"): Houses motor and gear system with innovative clutch mechanism that independently controls each leg's position
- Leg Modules: Feature ball hinges and clutch mechanisms for flexibility and motion range to navigate irregular surface topology
- Adaptable Feet Assembly: Interconnected triangular segments that conform to Psyche's uneven metallic terrain, maximizing contact area
- Tether Attachment Gambal: Flexible interface between anchoring system and space elevator ribbon
- System designed to handle height differences up to 10 meters and 34 degrees of tilt

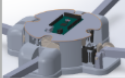
### References

- ASTM International. (2015). ASTM E955-15, Standard Test Method for "Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment."
- NASA. (2014). NASA-STD-5001, Structural Design and Test Factors of Safety for Spaceflight Hardware.
- NASA. (2016). NASA-STD-5016, Standard Materials and Processes Requirements for Spacecraft.
- NASA. (2014). NASA-STD-5017, Design and Development Requirements for Mechanisms.
- NASA. (2016). NASA-STD-5019, Fracture Control Requirements for Spaceflight Hardware.
- Viale, A., Carilli, M., & Molinari, C. R. (2021). Dynamics of a Nonrigid Orbital Siphon at a Near-Earth Asteroid. *Journal of Guidance, Control, and Dynamics*, 44(1), 137-148

## MIME-5504A



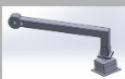
### Internal Component Images



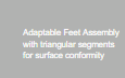
Central Hub ("Black Box") with motor and gear system featuring clutch mechanism



Tether Attachment Gambal interface to accommodate varying tension angles



Leg Modules with ball hinges and clutch mechanisms for flexible movement



Adaptable Feet Assembly with triangular segments for surface conformity

### Team Members:

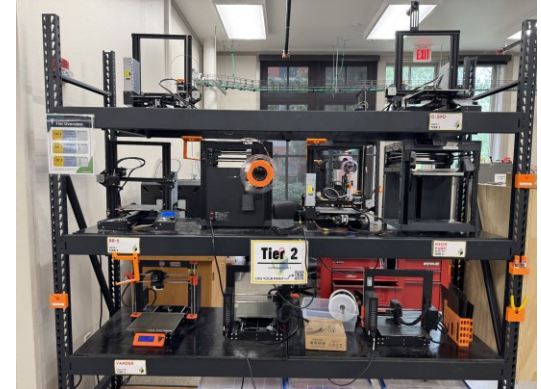
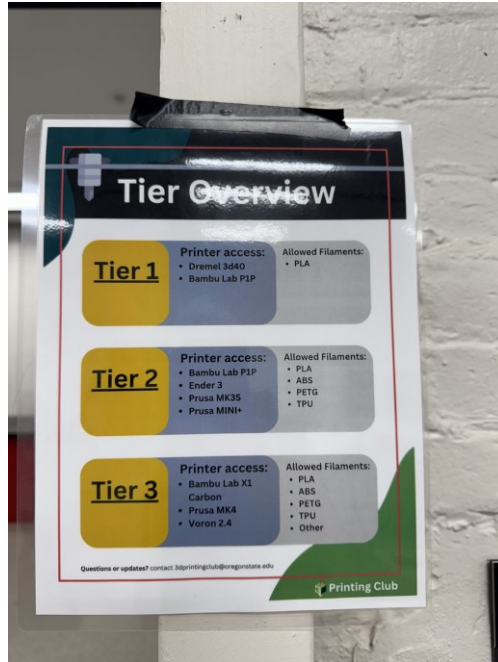
- Kiefer Gonsalves
- Kyle Ladda-Lewaren
- Jared Shahan
- Zachary Thomas
- Faculty Advisor: Sarah Oman
- Sponsor Mentor: Cassie Bowman (Arizona State University)

Kiefer

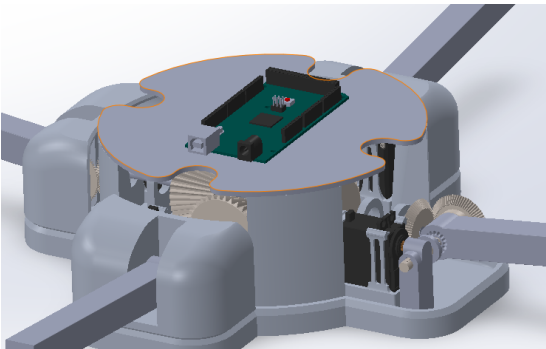
Upcoming Action Items:

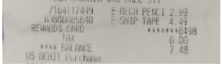
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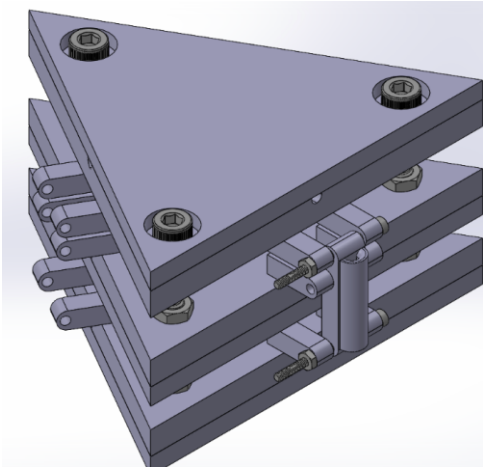
# iLabs 3D Printing - Training, Test Prints, & Assembly



# CAD Iterations with New Purchases



| Grant/ Funding Number (Index and Activity Code): FS034C ESNR |          |                                                                                                                                                                                                                                                         |                          |          |       |
|--------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|----------|-------|
| Bill of Materials                                            |          |                                                                                                                                                                                                                                                         |                          |          |       |
| Item Name                                                    | Price    | Receipt                                                                                                                                                                                                                                                 | Funding Summary:         |          |       |
| Magnets                                                      | \$9.59   | <br>Zilevez 400Pcs Small Magnets, Tiny Rare Earth Magnets, 4 Different Siz...<br>Sold by: Zilevez<br>Return or replace items: Eligible through March 6, 2025<br>\$9.59 | <b>Total Spend:</b>      |          |       |
| Packing Tape                                                 | \$4.49   |                                                                                                                                                                       |                          |          |       |
| PO1                                                          | \$292.86 | 4/7 Email: MIME Capstone Purchase - NASA Psyche Attachment Team - 5504 088                                                                                                                                                                              | <b>Budget:</b>           |          |       |
| PO2                                                          | \$85.39  | 4/11 Email: Round 2 of MIME Capstone Purchases                                                                                                                                                                                                          | <b>Remaining Budget:</b> |          |       |
|                                                              |          |                                                                                                                                                                                                                                                         |                          | \$392.33 | \$500 |
|                                                              |          |                                                                                                                                                                                                                                                         |                          |          | \$108 |



# How we Plan to Address Upcoming Engineering Challenges

- Electrostatic Gripper:
  - Continued research on high voltage systems and electroadhesion
  - How to simulate outside of trial and error
- Robotics & Controls:
  - Wiring and test
  - Integrating/Programming servo motors & microcontrollers
- 3D Print Quality:
  - Standardization of parameters & outlined order of operations
  - Making sure tasks are equally distributed
- Structure Failure Points:
  - Simulated FEA of parts interfaces under a cyclic temperature range and static loading schemes.
  - Explore part life expectancy through fatigue analysis

Thank you! Questions?